

Nephromyces Classifier Comparison: How the Choice of Reference Database Changes What You "See" in a Metagenome

Introduction

Nephromyces is a really interesting organism. It's an apicomplexan, which means it belongs to the same group as Plasmodium (the parasite that causes malaria) and Toxoplasma. But unlike most of its relatives, Nephromyces doesn't seem to harm its host. It lives inside the renal sac of molgulid tunicates and appears to have a stable commensal, possibly mutualistic, relationship with them. To make things even more unusual, Nephromyces itself harbors bacterial endosymbionts.

This complexity makes renal sac samples challenging to obtain. Sequencing captures a mix of host, Nephromyces, and bacterial DNA, creating a metagenome. Identifying what organisms are present is the main challenge.

The way most people can answer similar questions is to use a tool like Kraken2, which compares sequencing reads to a reference database and labels each read with the organism it best matches. The catch is that the answer you get depends a lot on which database you use. If your database doesn't contain references for protozoa, you can't detect them, even if they're in your sample. This project asks how big a deal that really is by running the same data through Kraken2 three times with three different databases and comparing the results.

The three databases compared are:

- **Basic:** bacteria, archaea, and viruses only. No eukaryotes at all.
- **PlusPF:** the standard Kraken2 build plus RefSeq protozoa and fungi.
- **PrackenDB:** a much broader build that includes GenBank and RefSeq assemblies of bacteria, archaea, protists, and fungi, plus the human genome and viral references.

The research question: How does the choice of database change what fraction of reads gets classified, what taxa appear to be present, and how much apicomplexan signal we can detect in a Nephromyces sample?

Project Goal: Develop a fully reproducible and automated Kraken2-based metagenomic analysis pipeline that can process multiple samples across different reference databases and compare how database composition influences taxonomic classification outcomes for downstream biological interpretation.

Methods

The pipeline runs on the URI/UMASS Unity supercluster. It takes a single FASTQ file (Oxford Nanopore long reads from a *Molgulid occidentalis* renal sac sample), runs Kraken2 once per database, and then processes the results with a series of Python scripts that convert raw classifier output into clean, comparable tables and figures.

Everything is controlled from a single config file ([config/config.yaml](#)), so the pipeline doesn't have any hardcoded paths. The main steps are:

1. **Run Kraken2** against each of the three databases. Each run produces a report listing the organisms detected and their percentages.
2. **Parse the reports** into Python ([01_parse_kraken.py](#)).
3. **Standardize the taxa** into a small set of biologically meaningful categories ([03_standardize_taxa.py](#)): Unclassified, Host/Chordata, Bacteria, Apicomplexa/Nephromyces-related, Other Eukaryotes, and Other. This step is what makes the three databases directly comparable.
4. **Compare the databases** ([04_compare_databases.py](#)) by computing pairwise differences in each category.
5. **Plot the results** ([05_plot_results.py](#)) to produce a classified-vs-unclassified bar chart and a stacked-bar plot of the full taxonomic composition.

The whole pipeline runs as a single SLURM job using [submit_pipeline.sh](#). For testing, a 10,000-read subset of the full dataset was used to confirm the pipeline runs end-to-end without errors.

Results

The comparison table shows really clear differences between the three databases:

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Category	Basic	PlusPF	PrackenDB
Unclassified	89.5%	85.3%	40.5%
Other	6.7%	3.7%	34.8%
Host/Chordata	0.0%	7.8%	0.5%
Apicomplexa/Nephromyces	0.0%	0.2%	0.6%
Bacteria	0.02%	0.0%	0.02%
Other Eukaryotes	0.0%	0.0%	0.0%

Table 1. Percentage of reads assigned to each taxonomic category by Kraken2 across three reference databases of differing scope. Values represent the proportion of total reads classified into each category for a 10,000-read subset of Oxford Nanopore data from a *Molgula occidentalis* renal sac sample. Categories were standardized via keyword matching on Kraken2 species-rank assignments.

The main finding is that database breadth dramatically affects the percentage of classified reads. With the basic database, nearly 90% of reads remain unclassified. PlusPF improves minimally to 85.3% unclassified. Only PrackenDB reduces unclassified reads to 40.5%, more than doubling classification. This stark contrast is clearly illustrated in Figure 1, emphasizing how database selection shapes the interpretation of metagenomic data.

The Apicomplexa/Nephromyces-related signal, which is what we actually care about for this project, also responds to the database choice. With the basic database, it's literally 0%, because there are no eukaryotic references. PlusPF picks up 0.2%, and PrackenDB picks up 0.6%, about three times higher. The numbers are small overall, but the trend is exactly what you'd expect: bigger database, more apicomplexan signal.

The Host/Chordata results are weird and worth pointing out. PlusPF says 7.8% host, but PrackenDB says only 0.5%. This is probably because PrackenDB has more diverse references, and reads that PlusPF assigned to host are getting "explained away" by other organisms once PrackenDB has more candidates to compare against. *Figure 2* shows the full taxonomic composition of all three databases side by side.

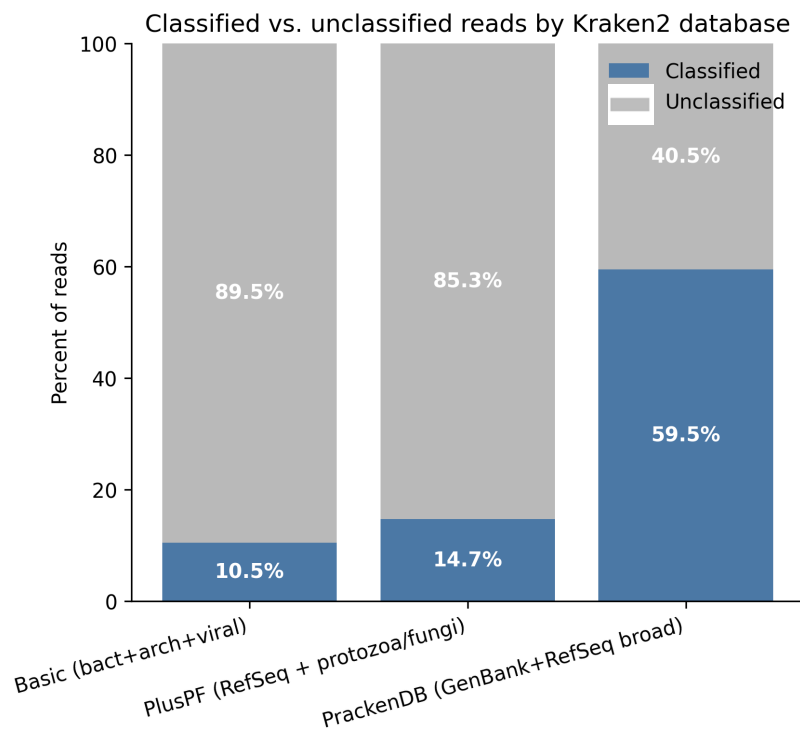


Figure 1. Percentage of reads classified versus left unclassified by Kraken2 across three reference databases. The unclassified fraction drops from 89.5% with the basic build (bacteria + archaea + viral) to 85.3% with PlusPF (RefSeq + protozoa/fungi) and 40.5% with PrackenDB (GenBank + RefSeq broad). Reads are from a 10,000-read subset of Oxford Nanopore sequencing of a *molgula occidentalis* renal sac sample.

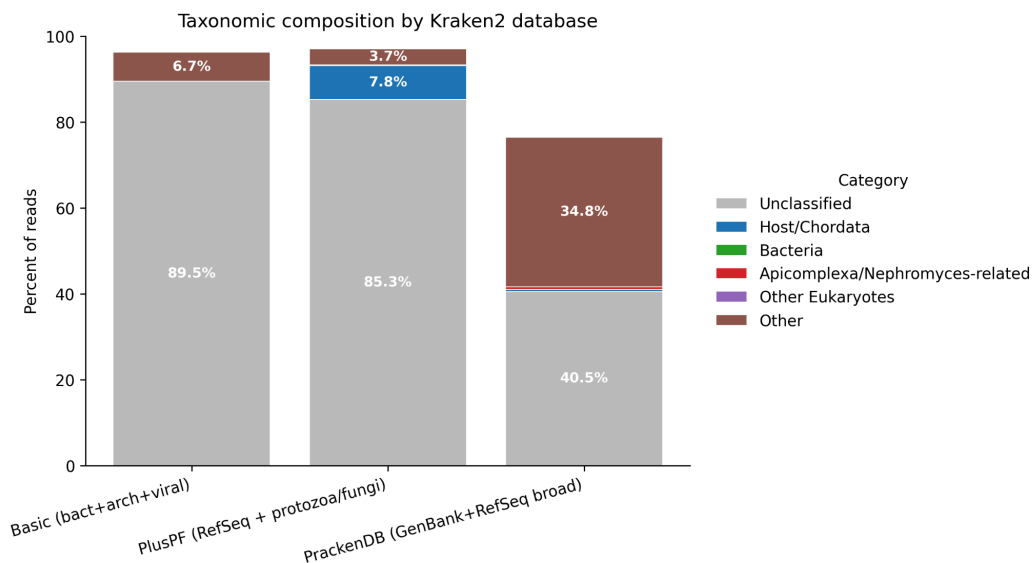


Figure 2. Taxonomic composition of the same 10,000-read subset under each of the three Kraken2 databases, with reads grouped into shared biological categories. Apicomplexa/Nephromyces-related signal is undetectable with the basic build, low with PlusPF (0.2%), and slightly higher with PrackenDB (0.6%). The "Other" category grows substantially with PrackenDB, reflecting the broader reference set capturing reads that the smaller databases leave unclassified.

Discussion

The most important takeaway is that unclassified reads should not automatically be interpreted as *Nephromyces* signal, just as classified reads should not be interpreted as complete representation of the sample. Unclassified reads may reflect sequencing artifacts, low-quality reads, host DNA absent from the reference database, or genuinely novel organisms. Accordingly, the pipeline avoids assuming that the 89.5% unclassified fraction observed in the Basic database represents biologically meaningful or novel content.

The second takeaway is that disagreement between databases doesn't mean any one of them is wrong. Each database has different references, so they're answering slightly different questions. PrackenDB's higher classification rate doesn't necessarily mean its assignments are more accurate, broader databases can also be more prone to false positives. But for an organism like *Nephromyces*, which is genuinely poorly represented in standard databases, going broader generally helps you see a signal that smaller databases miss entirely.

The main limitations are: (1) this run used only a 10,000-read test subset, so the exact numbers shouldn't be over-interpreted; (2) the standardization step uses keyword matching, which is robust but coarse; and (3) only three databases were compared, there are many others that could behave differently.

Conclusion

Running the same sample through different Kraken2 databases produces strikingly different outcomes. The unclassified fraction varies widely, from 40% to 90%, while detection of apicomplexan signals occurs only if the database actually contains apicomplexan references. The crucial insight for metagenomics is that conclusions depend strongly on database selection; presenting results from just one database can greatly overstate certainty about the sample.

Although PrackenDB shows the most apicomplexan signal of the three databases, the percentage is small (0.6%) and based on a relatively small test subset, so this analysis alone is not enough to confidently recommend any one database as the best choice for *Nephromyces* work. Next step will involve running the full dataset and confirming key apicomplexan assignments with a method like BLAST or read-level assembly which would help determine whether the trend observed here holds at scale.